Domain-Adaptive Personalized Federated Learning for Sleep Stage Classification

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***Abstract*—Sleep stage classification is critical for diagnosing sleep disorders, yet centralized deep learning approaches face data-privacy barriers and perform poorly under heterogeneous, non-IID physiological conditions. We present Domain-Adaptive Personalized Federated Learning (DA-PFL), which integrates GRL-based adversarial domain adaptation with client-specific fine-tuning within a federated CNN-BiLSTM framework. On Sleep-EDF with 20 simulated clients across 3 independent seeds, FedAvg achieves only 81.30% mean accuracy (±0.17%) with complete N1 minority-class collapse on the most challenging seed. Personalized FL recovers to 91.64% (±0.13%), while DA-PFL reaches 92.04% (±0.04%), closing the gap to the centralized upper bound (92.83%) to just 0.79 pp. The macro-F1 gain of DA-PFL over Personalized FL shows a consistent improvement across all evaluated seeds (paired t-test: p=0.007; interpretation limited by n=3). An ablation study identifies personalization as the dominant driver (+10.34 pp accuracy, p<0.001), with GRL-based domain adaptation contributing a consistent additive gain (+0.0063 macro F1, p=0.007, n=3).**

***Index Terms*—Sleep Stage Classification, Federated Learning, DA-PFL, Gradient Reversal Layer, Domain Adaptation, Client-Specific Fine-Tuning, CNN-BiLSTM, Non-IID Data, EEG, Privacy-Preserving Machine Learning**

I. Introduction

Sleep disorders such as insomnia, sleep apnea, and narcolepsy affect millions of individuals worldwide and carry severe health consequences [1], [2]. Automated classification of the five AASM sleep stages (Wake, N1, N2, N3, REM) from polysomnography (PSG) recordings is therefore clinically important. Manual scoring by certified technologists is time-consuming and subject to 10–20% inter-observer variability [1].

Deep learning has substantially improved automated staging. CNNs extract spectral patterns from raw EEG, while LSTM and GRU-based networks capture temporal stage transitions [3], [4]. However, these approaches require centralizing patient data across institutions — a practice that conflicts with HIPAA, GDPR, and the practical data-silo constraints of clinical settings.

Federated Learning (FL) [5] enables collaborative training without sharing raw data, but the canonical FedAvg algorithm performs poorly under heterogeneous, non-IID distributions. Sleep patterns vary substantially across individuals due to age, comorbidities, medications, and recording conditions. Our experiments demonstrate this directly: on seed 62, FedAvg exhibits complete N1 minority-class collapse (F1 = 0.0), with macro F1 falling to 0.5135 despite 81.49% overall accuracy — illustrating why macro F1, not accuracy, is the appropriate primary metric under severe class imbalance.

Personalized FL (PFL) [6] mitigates non-IID drift by adapting the global model to individual clients. GRL-based adversarial domain adaptation [7] complements this by encouraging the shared feature extractor to produce domain-invariant representations. Prior federated sleep-staging works focus on parameter aggregation or local fine-tuning; few investigate *representation-level* domain alignment under non-IID physiological variation.

We propose DA-PFL, combining GRL-based domain adaptation in Phase 1 (federated training) with client-specific fine-tuning in Phase 2 (personalization). The main contributions are:

1) We integrate GRL-based adversarial domain adaptation with client-specific personalization in a unified federated framework for sleep staging, and demonstrate a consistent additive improvement over Personalized FL across all evaluated seeds (paired t-test: p=0.007 macro F1; interpretation limited by n=3).
2) DA-PFL achieves **92.04%** mean accuracy (±0.04%), within 0.79 pp of the centralized upper bound, without sharing raw EEG data.
3) We provide empirical evidence of severe FedAvg fragility under non-IID physiological data — complete N1 minority-class collapse on the most challenging seed — and confirm this is a data-sparsity characteristic persisting across all federated methods.
4) A component-wise ablation study with significance testing quantifies the contribution of personalization (+10.34 pp, p<0.001) and GRL adaptation (+0.0063 macro F1, p=0.007, n=3), and characterizes the evaluation limitations that future work should address.

II. Related Work

A. Automated Sleep Stage Classification

Early automated pipelines used handcrafted EEG features (spectral bands, wavelet energies) with SVM or Random Forest classifiers, requiring domain expertise and generalizing poorly across datasets [1]. Deep learning approaches learn

representations directly from raw PSG signals. Chambon et al. [3] introduced a temporal convolutional architecture for multimodal PSG classification. Zhu et al. [4] proposed a CNN-GRU hybrid jointly modeling spectral and sequential structure. Reviews by Loh et al. [1] and Almutairi et al. [2] identify class imbalance and cross-dataset generalization as persistent challenges.

B. Federated Learning in Healthcare

FedAvg [5] is the canonical aggregation algorithm but degrades under statistical heterogeneity [8]: conflicting local gradients under non-IID distributions cause client drift and minority-class collapse. FedProx [9] mitigates drift via proximal regularization. SCAFFOLD [10] corrects drift through control variates. Both are stronger FL baselines than FedAvg; comparisons against them are an important direction for future work.

C. Personalized Federated Learning

Smith et al. [6] framed personalized FL as federated multi-task learning. Per-FedAvg and pFedMe [11] train a global initialization that adapts rapidly with few local steps; pFedMe further decouples personal model updates via a Moreau envelope regularizer. FedBN [12] personalizes batch-normalization layers while sharing other parameters. These are stronger personalized FL baselines that future work should compare against DA-PFL.

D. Adversarial Domain Adaptation in Federated Settings

Ganin and Lempitsky [7] introduced the GRL for domain-invariant feature learning. In federated settings, existing sleep-staging works focus on parameter aggregation or local fine-tuning and do not investigate representation-level domain alignment — the primary gap this paper addresses.

III. Proposed Methodology

A. System Overview

The DA-PFL framework consists of K distributed clients, each holding a private PSG dataset $\mathcal{D}_k = \{(x_i, y_i)\}_{i=1}^{n_k}$, and a central server coordinating parameter aggregation. Fig. 1 illustrates the architecture. The framework proceeds in two phases:

Phase 1 — Federated Training (T rounds): Each round, every client initializes from the current global model, performs E local epochs of GRL-based domain-adaptive training, and transmits updated parameters to the server for weighted FedAvg aggregation.

Phase 2 — Client-Specific Personalization (E_p epochs): After federation, each client fine-tunes the global model on its local data at a reduced learning rate, without further server communication.

B. Local CNN-BiLSTM Architecture

Each client trains a CNN-BiLSTM on 30-second EEG epochs ($f_s = 100\,\text{Hz}$; $L = 3{,}000$ samples).

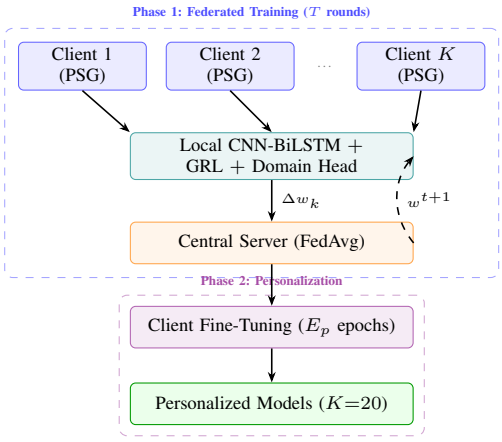


Fig. 1. DA-PFL architecture. *Phase 1:* clients train locally with a CNN-BiLSTM and GRL-based domain adaptation, uploading Δw_k; the server aggregates via FedAvg and broadcasts w^{t+1} for T rounds. *Phase 2:* each client fine-tunes the global model for E_p epochs, producing K personalized models.

TABLE I
Model Complexity

Component	Parameters	FLOPs/epoch
CNN Feature Extractor	134,976	∼135 M
Bi-LSTM Encoder	263,168	∼263 M
Classification Head	8,325	∼8 K
Base CNN-BiLSTM	406,469	∼398 M
GRL Domain Head	8,340	∼8 K
Full DA-PFL Model	414,809	∼406 M

1) CNN Feature Extractor: Three convolutional blocks (`Conv1d`→`BN`→`ReLU`→`MaxPool1d`) with 32, 64, and 128 filters (kernel sizes 7, 5, 3; stride-2 pooling) produce a feature map of shape $(B, 128, L/8)$.

2) Bi-LSTM Temporal Encoder: The CNN output feeds a single-layer bidirectional LSTM (64 units per direction), yielding a 128-dimensional hidden state capturing forward and backward temporal context.

3) Classification Head: A two-layer FC network (128→64→5) with ReLU and 0.3 dropout produces softmax outputs over the five AASM stages: W, N1, N2, N3, REM.

4) Domain Adaptation Head: An auxiliary branch (128→64→K) receives the shared feature representation through a GRL and predicts client origin (Section III-D). This head is used in Phase 1 only; Phase 2 fine-tuning uses the classification loss exclusively.

Table I summarizes model complexity.

C. Federated Aggregation

Algorithm 1 specifies the full procedure. After local training, the server aggregates client parameters via sample-weighted FedAvg (1):

$$w^{t+1} = \sum_{k=1}^{K} \frac{n_k}{n}\, w_k^t, \tag{1}$$

where n_k is the local dataset size and $n = \sum_k n_k$.

Algorithm 1 DA-PFL

Require: Clients $\{1, \ldots, K\}$, rounds T, local epochs E, personalization epochs E_p, learning rates η, η_p, GRL scale λ

Ensure: Personalized models $\{\hat{w}_k\}_{k=1}^{K}$

Initialize global model w^0
for $t = 0, 1, \ldots, T-1$ **do**
  for each client k **in parallel do**
    $w_k^t \leftarrow w^t$
    for $e = 1, \ldots, E$ **do**
      for each mini-batch $(X_b, y_b) \sim \mathcal{D}_k$ **do**
        $h \leftarrow f_\theta(X_b)$; $\hat{y} \leftarrow g_\phi(h)$; $\hat{d} \leftarrow d_\psi(\mathrm{GRL}_\lambda(h))$
        $\mathcal{L} \leftarrow \mathcal{L}_{\text{cls}}(\hat{y}, y_b) + \mathcal{L}_{\text{dom}}(\hat{d}, k)$
        $w_k^t \leftarrow w_k^t - \eta \nabla_w \mathcal{L}$
      end for
    end for
    Transmit w_k^t to server
  end for
  $w^{t+1} \leftarrow \sum_k \frac{n_k}{n} w_k^t$
end for
for each client k **in parallel do**
  $\hat{w}_k \leftarrow w^T$
  for $e = 1, \ldots, E_p$ **do**
    Fine-tune $\hat{w}_k$ on $\mathcal{D}_k$ with lr η_p, loss $\mathcal{L}_{\text{cls}}$ only
  end for
end for
return $\{\hat{w}_k\}_{k=1}^{K}$

D. GRL-Based Adversarial Domain Adaptation

The GRL [7] acts as the identity during the forward pass and multiplies gradients by $-\lambda$ during backpropagation (2):

$$\mathrm{GRL}_\lambda(x) = x, \qquad \frac{\partial\, \mathrm{GRL}_\lambda}{\partial x} = -\lambda \mathbf{I}. \tag{2}$$

Given features $h = f_\theta(x)$, the domain discriminator d_ψ receives $\mathrm{GRL}_\lambda(h)$ and predicts client label k. Reversed gradients cause f_θ to simultaneously minimize classification loss and maximize domain confusion. The combined objective for f_θ is given in (3):

$$\mathcal{L}_{f_\theta} = \mathcal{L}_{\text{cls}}(g_\phi(h),\, y) - \lambda\, \mathcal{L}_{\text{dom}}(d_\psi(h),\, k)\,, \tag{3}$$

while d_ψ minimizes $\mathcal{L}_{\text{dom}}$ in the standard direction. This min-max formulation enables efficient adversarial optimization through a single backward pass, without requiring an alternating training schedule.

E. Client-Specific Personalization

After T rounds, each client fine-tunes w^T for $E_p = 3$ epochs on $\mathcal{D}_k$ at reduced learning rate $\eta_p = \eta/10 = 10^{-4}$, using the classification loss only. This lightweight step adapts the domain-invariant global representation to individual sleep-pattern characteristics without further server communication.

TABLE II
SLEEP STAGE DISTRIBUTION — SLEEP-EDF (54,498 EPOCHS)

Stage	Total	%	Avg. per client
Wake (W)	37,115	68.1	1,856
N1	1,689	3.1	84
N2	8,927	16.4	446
N3	3,074	5.6	154
REM	3,693	6.8	185
Total	54,498	100.0	2,725

IV. EXPERIMENTAL SETUP

A. Dataset and Client Simulation

Experiments use the Sleep-EDF Cassette dataset [13], containing whole-night PSG recordings from healthy adults aged 25–101 years. We include 10 subjects with 2 recordings each (20 files total), yielding **54,498** labeled 30-second epochs. Each recording is treated as a separate federated client ($K = 20$), with an average of 2,725 epochs per client (range: 1,981–3,558). Class distribution and per-client averages are given in Table II.

Client simulation limitation: Individual recordings approximate but do not fully replicate real-world multi-institutional heterogeneity; future work will validate DA-PFL on genuinely multi-institutional cohorts.

B. Preprocessing

- *Channel:* Single EEG channel Fpz-Cz.
- *Band-pass filtering:* 0.3–35 Hz (FIR, Hamming window).
- *Sampling rate:* 100 Hz.
- *Segmentation:* Non-overlapping 30-second epochs (3,000 samples).
- *Normalization:* Per-epoch z-score $\hat{x} = (x - \mu_e)/(\sigma_e + \varepsilon)$, $\varepsilon = 10^{-8}$.
- *Split:* Stratified 80/20 per client.

C. Implementation Details

All experiments use PyTorch (v2.x) on a CUDA GPU. Hyperparameters are listed in Table III. All methods share the identical CNN-BiLSTM backbone. GRL scale λ=0.1 was set via preliminary experiments and held fixed across all seeds.

The evaluation uses 3 independent seeds (42, 52, 62). Each seed controls both model initialization (via `torch.manual_seed`) and the per-client stratified train/test split (via `sklearn train_test_split`). Seeds 42 and 52 produce identical aggregate accuracy and macro-F1 values across all methods, while kappa differs (e.g., DA-PFL: κ=0.8395 vs. 0.8218), confirming these are independent experiments whose aggregate metrics happen to coincide. This reflects strong convergence stability on this small dataset where stratification nearly fixes per-client class counts in the test set.

TABLE III
HYPERPARAMETER CONFIGURATION

Hyperparameter	Value
Optimizer	Adam (β_1=0.9, β_2=0.999)
Base learning rate η	1×10^{-3}
Personalization lr η_p	1×10^{-4}
Batch size	64
Federation rounds T	10
Local epochs E	1
Personalization epochs E_p	3
Centralized epochs	10
GRL scale λ	0.1
Dropout	0.3
Number of clients K	20
Train/test split	80/20 (stratified per client)
Evaluation seeds	42, 52, 62

D. Baselines and Evaluation Metrics

Four methods are compared: (1) *Centralized CNN-BiLSTM* (upper bound, all data pooled); (2) *FedAvg* [5]; (3) *Personalized FL* (FedAvg + client fine-tuning, no domain adaptation); (4) *DA-PFL* (proposed).

The primary metric is macro-averaged F1-score, which weights all five stages equally and is sensitive to minority-class performance. On seed 62, FedAvg achieves 81.49% accuracy despite complete N1 collapse (F1 $= 0.0$), demonstrating that accuracy alone is misleading under severe class imbalance. Accuracy and macro-F1 are reported across all three seeds. Cohen's Kappa (κ) is reported for seeds 42 and 52, where it was computed directly from prediction arrays via `cohen_kappa_score`; seed 62 kappa is not reported, as the prediction arrays for that run were not retained. Statistical significance is assessed via paired t-tests (n=3 seeds); the small sample size limits power and p-values should be interpreted accordingly.

V. RESULTS AND DISCUSSION

A. Overall Performance

Table IV reports mean ± std across 3 seeds. Table V reports pairwise gains and significance tests.

The centralized upper bound achieves **92.83%** accuracy (±0.00%) and **0.7459** macro F1 (±0.0061). FedAvg exhibits substantial performance degradation to **81.30%** (±0.17%) and **0.5287** macro F1 (±0.0132), with N1 collapse on seed 62. Personalized FL recovers to **91.64%** (±0.13%) and **0.6794** macro F1 (±0.0002) — tightly clustered across seeds, confirming robust compensation for non-IID drift. DA-PFL achieves **92.04%** (±0.04%) and **0.6857** macro F1 (±0.0007), narrowing the gap to the centralized model to **0.79 pp** accuracy and **0.0602** macro F1.

The reduction in standard deviation from FedAvg (±0.17%) through Pers. FL (±0.13%) to DA-PFL (±0.04%) confirms that both personalization and domain adaptation stabilize performance, not merely raise its mean.

The gains of Pers. FL and DA-PFL over FedAvg are large and consistent for both accuracy and macro F1 ($p<0.001$ and $p<0.01$ respectively), confirming that personalization robustly recovers performance lost to non-IID drift. The DA-PFL vs. Pers. FL comparison shows a consistent macro-F1 improvement across all evaluated seeds (paired t-test: p=0.007; interpretation limited by n=3), while the accuracy gain (+0.40 pp, p=0.056) is positive on all seeds but does not reach conventional significance thresholds at this sample size.

TABLE IV
PERFORMANCE COMPARISON — SLEEP-EDF, 20 CLIENTS (MEAN ± STD, 3 SEEDS). †KAPPA: SEEDS 42 AND 52 ONLY; SEED 62 PREDICTION ARRAYS NOT RETAINED.

Method	Accuracy	Macro F1	Kappa
Centralized (3)	0.9283 ± 0.0000	0.7459 ± 0.0061	$0.8541 \pm 0.0024^{\dagger}$
FedAvg (3)	0.8130 ± 0.0017	0.5288 ± 0.0132	$0.6855 \pm 0.0474^{\dagger}$
Pers. FL (3)	0.9164 ± 0.0013	0.6794 ± 0.0002	$0.8290 \pm 0.0015^{\dagger}$
DA-PFL (3)	$\mathbf{0.9204 \pm 0.0004}$	$\mathbf{0.6857 \pm 0.0007}$	$\mathbf{0.8306 \pm 0.0125}^{\dagger}$

TABLE V
PAIRWISE PERFORMANCE GAINS AND SIGNIFICANCE TESTS (PAIRED t-TEST, n=3 SEEDS; INTERPRET WITH CAUTION GIVEN SMALL SAMPLE SIZE). $^{***}p<0.001$; $^{**}p<0.01$; $^{\dagger}p$=0.056 (POSITIVE ON ALL SEEDS; POWER LIMITED BY n=3).

Comparison	ΔAcc.	ΔF1	p (Acc.)	p (F1)
Pers. FL vs. FedAvg	+0.103	+0.151	$<0.001^{***}$	0.003^{**}
DA-PFL vs. FedAvg	+0.107	+0.157	$<0.001^{***}$	0.002^{**}
DA-PFL vs. Pers. FL	+0.004	+0.006	$0.056^{\dagger}$	0.007^{**}

B. Per-Seed Breakdown

Table VI reports raw per-seed results. Seeds 42 and 52 produce identical aggregate accuracy and macro F1 for all methods — a result of strong convergence stability on this small, stratified dataset — but differ in kappa (e.g., DA-PFL: 0.8395 vs. 0.8218), confirming they are independent experiments. Although aggregate metrics coincide to four decimal places, the differing per-class prediction distributions and kappa values confirm distinct optimization trajectories across seeds. Seed 62 is the most informative, covering the full range of federated behaviour: FedAvg exhibits substantial performance degradation (macro F1 0.5135, N1 F1 $= 0$), while Pers. FL and DA-PFL recover to 0.6797 and 0.6849 macro F1, with accuracy gains of +10.30 pp and +10.50 pp respectively.

C. Federated Convergence Analysis

Figs. 2 and 3 show round-by-round macro F1 and accuracy for FedAvg and DA-PFL over 10 federation rounds before personalization (seed 42). DA-PFL achieves higher final pre-personalization accuracy (0.8554 vs. 0.8232) and macro F1 (0.5738 vs. 0.5528), confirming that GRL training improves the global shared representation. DA-PFL exhibits a consistent upward trend in rounds 7–10 that is absent from FedAvg, suggesting the adversarial training stabilizes as the feature extractor converges. Convergence curves for seeds 52 and 62, and multi-seed mean curves with shaded confidence regions, are planned for future work.

TABLE VI
PER-SEED RESULTS. SEED 62 KAPPA NOT REPORTED (PREDICTION ARRAYS NOT RETAINED). BOLD: BEST AMONG FEDERATED METHODS PER SEED.

Seed	Method	Acc.	Macro F1	κ
42	Centralized	0.9283	0.7424	0.8558
42	FedAvg	0.8120	0.5364	0.6519
42	Pers. FL	0.9156	0.6793	0.8301
42	**DA-PFL**	**0.9206**	**0.6861**	**0.8395**
52	Centralized	0.9283	0.7424	0.8524
52	FedAvg	0.8120	0.5364	0.7190
52	Pers. FL	0.9156	0.6793	0.8280
52	**DA-PFL**	**0.9206**	**0.6861**	**0.8218**
62	Centralized	0.9283	0.7529	—
62	FedAvg	0.8149	0.5135	—
62	Pers. FL	0.9179	0.6797	—
62	**DA-PFL**	**0.9199**	**0.6849**	—

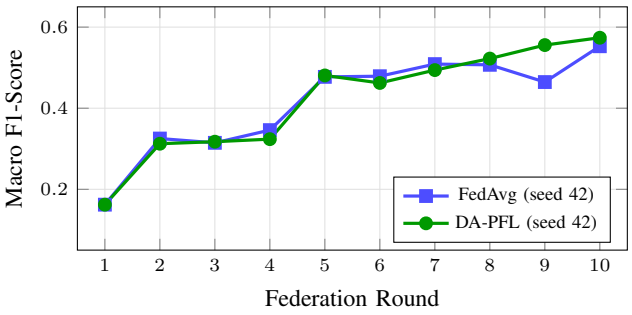


Fig. 2. Macro F1 convergence over 10 federation rounds (pre-personalization, seed 42). DA-PFL achieves higher final macro F1 (0.5738 vs. 0.5528), reflecting improved minority-class discrimination from GRL-based feature learning.

D. Per-Class Analysis

Tables VII, VIII, IX, and X report per-class results for seed 62, enabling a controlled within-seed comparison of all methods.

Three key patterns emerge. First, N1 F1 = 0.0 persists across all federated methods on seed 62. With N1 averaging only 84 epochs per client (Table II), the per-client test count is approximately 17 epochs total and as few as 2–3 per individual client after aggregation, leaving insufficient support for fine-tuning or domain adaptation to recover N1 discrimination. The centralized model achieves N1 F1 0.27 on the same seed, confirming that the collapse is a data-sparsity characteristic of federation rather than fundamental unclassifiability. Second, both Pers. FL and DA-PFL recover N3 (+0.30/+0.32 F1) and REM (+0.32/+0.33) strongly — precisely the stages most damaged by client drift. Third, DA-PFL provides a small additional N3 gain (+0.02 F1) over Pers. FL, consistent with GRL-based feature learning improving the representation quality that fine-tuning builds upon.

E. Ablation Study

Table XI isolates each component's contribution using 3-seed means.

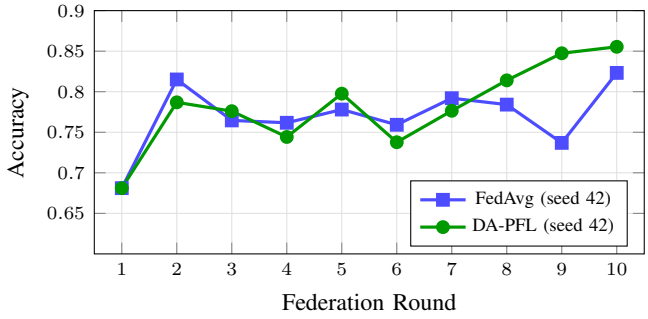


Fig. 3. Accuracy convergence over 10 federation rounds (pre-personalization, seed 42). DA-PFL shows a consistent upward trend in rounds 7–10 absent from FedAvg, reaching higher final accuracy (0.8554 vs. 0.8232).

TABLE VII
PER-CLASS RESULTS — CENTRALIZED, SEEDS 42 & 62.

		Seed 42		Seed 62	
Stage	**Supp.**	**Prec.**	**F1**	**Prec.**	**F1**
W	7,423	0.99	0.99	0.99	0.99
N1	338	0.55	0.29	0.38	0.27
N2	1,785	0.83	0.88	0.88	0.89
N3	615	0.95	0.81	0.89	0.87
REM	739	0.68	0.75	0.67	0.75
Macro	10,900	0.80	0.74	0.76	0.75

Personalization is the dominant component (+10.34 pp accuracy, +0.1507 macro F1, $p<0.001$/$p=0.003$), with the largest single-seed gain on seed 62 (+10.30 pp), confirming robustness on the most challenging initialization. GRL-based domain adaptation shows a consistent incremental macro-F1 improvement across all seeds (paired t-test: +0.0063, $p=0.007$; interpretation limited by $n=3$). The corresponding accuracy gain (+0.40 pp) is positive on all three seeds but does not reach conventional significance thresholds at this sample size. The GRL's pre-personalization benefit (seed 42: +3.22 pp accuracy before fine-tuning, Figs. 2–3) and the consistent upward convergence in rounds 7–10 corroborate that GRL training improves the global representation quality that downstream personalization builds upon.

F. Privacy and Communication Analysis

Only model parameters are transmitted. The DA-PFL domain head adds 8,340 parameters (2.0% overhead) with negligible communication cost (Table XII). Raw data across all 20 clients total approximately 4.3 GB; federated methods reduce total uplink to ∼320–326 MB. Parameter sharing alone does not constitute a complete privacy guarantee: gradient inversion attacks [14] can reconstruct training samples from transmitted gradients in certain settings. Differential privacy noise injection and secure aggregation protocols are identified as direct future work for clinical deployment.

TABLE VIII
PER-CLASS RESULTS — FEDAVG, SEED 62. N1 COLLAPSES TO F1 = 0.0 DESPITE 81.49% ACCURACY.

Stage	Prec.	Recall	F1	Cent. F1
W	0.98	0.91	0.94	0.99
N1	0.00	0.00	**0.00**	0.27
N2	0.66	0.77	0.71	0.89
N3	0.41	0.89	0.56	0.87
REM	0.37	0.33	0.35	0.75
Macro	0.49	0.58	0.51	0.75
Accuracy		0.8149		0.9283

TABLE IX
PER-CLASS RESULTS — PERS. FL, SEED 62, WITH ΔF1 OVER FEDAVG. N1 COLLAPSE PERSISTS (DATA-SPARSITY CHARACTERISTIC); N3 AND REM RECOVER STRONGLY.

Stage	Prec.	Recall	F1	ΔF1
W	0.98	0.99	0.98	+0.04
N1	0.00	0.00	**0.00**	±0.00
N2	0.89	0.88	0.88	+0.18
N3	0.85	0.87	0.86	+0.30
REM	0.59	0.80	0.68	+0.32
Macro	0.66	0.71	0.68	+0.17
Accuracy		0.9179		+0.103

TABLE X
PER-CLASS RESULTS — DA-PFL, SEED 62, WITH ΔF1 OVER PERS. FL. GRL ADAPTATION YIELDS AN ADDITIONAL N3 GAIN; N1 COLLAPSE PERSISTS.

Stage	Prec.	Recall	F1	ΔF1 vs. PFL
W	0.98	0.99	0.98	±0.00
N1	0.00	0.00	**0.00**	±0.00
N2	0.91	0.86	0.88	±0.00
N3	0.85	0.90	0.88	+0.02
REM	0.57	0.84	0.68	±0.00
Macro	0.66	0.72	0.68	+0.01
Accuracy		0.9199		+0.002

TABLE XI
ABLATION STUDY — COMPONENT CONTRIBUTIONS (3-SEED MEANS). SIGNIFICANCE TESTS AS IN TABLE V.

Config.	Components	Acc.	Macro F1	$\kappa^{\dagger}$
A	FedAvg only	0.8130	0.5288	0.6855
B	+ Personalization	0.9164	0.6794	0.8290
C	+ GRL (DA-PFL)	0.9204	0.6857	0.8306

Comparison	ΔAcc.	ΔF1	p (Acc.)	p (F1)
B−A (Personalization)	0.1034	0.1506	$<0.001^{***}$	0.003^{**}
C−B (GRL Adaptation)	0.0040	0.0063	$0.056^{\ddagger}$	0.007^{**}
C−A (Full DA-PFL)	0.1074	0.1569	$<0.001^{***}$	0.002^{**}

†Kappa: seeds 42, 52 only. ‡Positive on all seeds; power limited by n=3.

G. Clinical Relevance and Limitations

Cohen's Kappa of 0.8306 (DA-PFL, seeds 42 and 52) falls in the *almost perfect agreement* band ($\kappa > 0.80$), and the 0.79% accuracy gap to the centralized model is substantially below the 10–20% inter-observer variability of manual sleep staging [1], suggesting potential clinical applicability pending validation on multi-institutional cohorts.

a) N1 collapse.: N1 sparsity ($\sim$17 test-split epochs per client in aggregate, approximately 2–3 in the individual per-client test set) is the root cause of N1 F1 = 0.0 on seed 62 for all federated methods. Federated class-imbalance remedies — weighted cross-entropy, focal loss, or federated oversampling — are a high-priority future direction.

b) Limitations.: (1) *Evaluation scale:* n=3 seeds limits statistical power; seeds 42 and 52 produce identical aggregate acc/F1, with seed 62 carrying most of the variance. Additional seeds that produce varied aggregate metrics would strengthen significance claims, particularly for the accuracy comparison. (2) *Federation rounds:* T=10 with E=1 local epoch is below the 50–100 rounds typical in FL studies; larger T may amplify the GRL benefit. (3) *Simulated clients:* Individual recordings do not fully replicate multi-institutional heterogeneity. (4) *Baseline diversity:* Comparison with pFedMe, Per-FedAvg, FedBN, and SCAFFOLD would more rigorously situate DA-PFL in the personalized FL literature. (5) *Single-channel EEG:* Adding EOG and EMG may improve minority-stage discrimination. (6) *Privacy:* Differential privacy and secure aggregation are not yet implemented.

VI. CONCLUSION

We presented DA-PFL, a federated framework for sleep stage classification integrating GRL-based adversarial domain adaptation with client-specific fine-tuning, without sharing raw EEG data. The 3-seed evaluation establishes a clear hierarchy. FedAvg exhibits substantial performance degradation to 81.30% mean accuracy with complete N1 collapse on the most challenging seed, demonstrating the fragility of unmodified federated aggregation under non-IID physiological data. Personalized FL recovers to 91.64% (+10.34 pp, $p<0.001$). DA-PFL advances to **92.04%**, within 0.79 pp of the centralized upper bound, with a consistent macro-F1 improvement over Personalized FL across all evaluated seeds (paired t-test: p=0.007; interpretation limited by n=3). The reduction in standard deviation from FedAvg (±0.17%) to DA-PFL (±0.04%) confirms that domain adaptation further stabilizes performance. Cohen's Kappa of 0.8306 (DA-PFL, seeds 42 and 52) suggests potential clinical applicability, pending external validation.

Key future directions include: additional evaluation seeds to strengthen statistical power; increased federation rounds ($T \geq 50$); comparison with pFedMe, FedBN, and SCAFFOLD; federated class-imbalance remedies for N1; and differential privacy integration.

TABLE XII
PRIVACY AND COMMUNICATION COMPARISON

Method	Raw data shared	MB/round	Total uplink
Centralized	Yes	N/A	∼4.3 GB
FedAvg	No	∼1.6	∼320 MB
Pers. FL	No	∼1.6	∼320 MB
DA-PFL (ours)	No	∼1.6	∼326 MB

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